

ICT AND SOCIETY

1. THE CURRENT STATE OF ICT

The Fourth Industrial Revolution (Industry 4.0) We are currently living in an era defined by the convergence of digital, biological, and physical innovations. ICT is no longer just a tool for work but an integrated part of our existence.

Core Technologies Driving This Era:

- **5G Networks:** Provides ultra-fast connectivity.
- **Artificial Intelligence (AI):** Machines that simulate human intelligence (e.g., ChatGPT, Siri).
- **Internet of Things (IoT):** Everyday objects connected to the internet (e.g., smart fridges, connected cars).
- **Cloud Computing:** Storing and accessing data over the internet instead of local hard drives.
- **Big Data:** The ability to analyze massive amounts of information to predict trends.

Evolution of Devices

- **Past:** Computers were stationary devices confined to desks.
- **Present:** ICT is **ubiquitous** (present everywhere)—embedded in watches, cars, appliances, and city infrastructure.

Smart Environments The trend is shifting toward automated systems where devices communicate without human intervention.

- **Smart Cities:** Using sensors to manage traffic, waste, and energy efficiently.
- **Smart Homes:** Automated lighting, heating, and security systems.
- **Economic Shift:** Transition from a simple "Information Age" to a "**Digital Economy**" (where data is considered the "new oil").

PART A. ICT AND SOCIETY

2. IMPACT OF ICT IN OUR DAILY LIVES

ICT has fundamentally altered daily routines, making tasks more convenient, faster, and cost-effective.

A. Communication & Social Interaction

- **Shift:** Replaced traditional mail/landlines with Instant Messaging and Video Conferencing.

- **Tools:** Messenger, WhatsApp, Zoom, Discord.
- **Benefit:** Real-time connection regardless of distance; families stay connected globally.
- **Drawback: FOMO** (Fear Of Missing Out) and reduced face-to-face interaction, leading to social isolation.

B. Financial & Commercial Convenience

- **Digitized Errands:** E-commerce platforms (Amazon, Shopee, Lazada) allow shopping from home 24/7.
- **Cashless Society:** Mobile banking and digital wallets (GCash, Maya, Apple Pay) make transactions instant and trackable.
- **Impact:** Saves significant time and energy on paying bills, booking flights, and ordering food.

C. Access to Information & Entertainment

- **Democratized Information:** Search engines (Google) replaced physical encyclopedias; answers are available in seconds.
- **On-Demand Entertainment:** Streaming services (Netflix, Spotify, YouTube) replaced traditional cable TV and radio, allowing users to choose *what* to watch and *when*.

3. IMPACT OF ICT IN SOCIETY

Beyond individual convenience, ICT drives large-scale structural changes in how major sectors of society function.

A. Business and Commerce

- **The Gig Economy:** Platforms like Upwork, Uber, and Grab create flexible freelance jobs, moving away from traditional 9-to-5 employment.
- **Global Market Access:** Small businesses can reach international customers through social media marketing and online stores.
- **Automation:** Use of software to handle repetitive tasks (data entry, manufacturing), increasing efficiency but risking job displacement.

B. Government and Governance (E-Governance)

- **Transparency & Efficiency:** Governments use ICT to deliver services online (e.g., renewing passports, filing taxes, national ID systems).

- **Public Safety:** Use of surveillance cameras and data analysis to monitor crime and traffic (Command Centers).
- **Citizen Engagement:** Social media allows citizens to report issues directly to government agencies.

C. Education (Transformation)

- **Shift:** From **Teacher-Centered** to **Learner-Centered**.
- **Tools:** E-learning platforms (Canvas, Blackboard), digital textbooks, virtual classrooms.
- **Distance Learning:** Became the lifeline during the COVID-19 pandemic; allows remote students to access high-quality resources previously unavailable to them.

D. Healthcare (e-Health)

- **Efficiency: Electronic Health Records (EHR)** allow instant access to patient history across different hospitals, reducing medical errors.
- **Telemedicine:** Enables doctor consultations via video calls, vital for rural areas without specialists.
- **Wearable Tech:** Smartwatches monitor heart rate and sleep in real-time for **preventative care**.

4. CHALLENGES AND NEGATIVE IMPACTS

While ICT brings progress, it introduces new societal problems that must be managed.

A. The Digital Divide

- **Definition:** The gap between those who have access to the internet/modern devices and those who do not.
- **Consequence:** Creates inequality. Students without internet fall behind in school; workers without digital skills cannot find jobs.

B. Environmental Impact

- **E-Waste (Electronic Waste):** Discarded computers and phones contain toxic materials (lead, mercury) that harm the environment if not recycled properly.
- **Energy Consumption:** Massive data centers required for Cloud Computing and AI consume huge amounts of electricity.

C. Health and Psychological Issues

- **Physical Health:** Prolonged screen time leads to eye strain, poor posture, and sedentary lifestyles (obesity).
- **Mental Health:** Cyberbullying, internet addiction, and anxiety caused by social media comparisons.

D. Security and Privacy Threats

- **Cybercrime:** Identity theft, phishing scams, and hacking.
- **Misinformation:** The rapid spread of "Fake News" can destabilize communities and influence public opinion (e.g., during elections).